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Design Areas - CS1060 Homework 2

Disclaimer:

Yes, I used AI to create the polished breakdown seen below, but I wrote the full MVP description of this app myself. I included my own list of stakeholders, clarified the user experience, and emphasized the value proposition on my own. The creative heavy lifting was done by me, I just used ChatGPT to condense and structure it efficiently.

oArena - internet-enabled mobile rowing app

Idea & Differentiation

- **The Idea:**

oArena is a mobile app that connects and interfaces with the industry-standard Concept2 rowing machine via Bluetooth. It enables users to:

- Launch solo workouts with customized parameters.
- Create, join, invite friends to, and participate in **real-time, gamified rowing races/workouts** with other users, regardless of location.
- Track live workout metrics, and maintain a personal workout history, along with trends in performance.

- **Existing Solutions:**

- *ErgData* → Concept2's proprietary ergometer mobile app. Simplifies workout setup and tracks data, but does not support real-time competition or a gamified experience.
- *Ergatta* → Offers gamified rowing workouts with others, but requires buying their proprietary machine and using their software (not accessible to most, and not a familiar experience for rowers). Rowers tend to hate using non-C2 ergs, and most gyms opt for C2's anyway.
- *Rowmetrics* → Has experimented with live racing, but lacks mass adoption and seamless mobile-first usability. It was clunky, required a USB cable to use, and

was only available through a website. It was ahead of its time, before mobile technology was able to support a seamless and lightweight app with an intuitive and fun UI.

- *Zwift* → Massive success in the cycling sphere. Rowers are just as fanatical as cyclists, if not more so, so it would be interesting to see if rowers would lean into a similar experience to the same extent.

- **Differentiation:**

- Arena uniquely combines:

- **Machine-agnostic compatibility** – works on the industry-standard Concept2 PM5 erg monitor without requiring a new hardware purchase/firmware installation. Rowers can train familiarly, and just add this on top of their usual experience.
 - **Mobile-native design** – lowers friction; users can set up and race from their phone. Ideally, setting up a workout through the app should be quicker than doing it through the ergometer itself.
 - **Real-time competitive racing** – the core differentiator; not just asynchronous leaderboards, but live “head-to-head” matches across distance. Training remotely usually limits your competitive experience to comparing workout time after the fact - a live experience mimics racing on the water, which most rowers would agree is the best part of the sport.

- **Why Now (Feasibility):**

- **Bluetooth firmware improvements** on Concept2 PM5 monitors make reliable device-app connections possible. This is more reliable than previous PM3 and PM4 monitor generations.
 - **Broad user base** – Concept2 is the universal erg in rowing, CrossFit, and gyms, creating a natural, built-in install base. More gyms are realizing that Concept2 is the rowers’ choice. This makes it easier for non-rowers to pick up the sport, because they can expect the same rowing experience at every gym. Concept2 is also growing steadily YoY, and the likes of Instagram and Tiktok are exposing more people to rowing every day. More people own their own C2’s than ever before.
 - **Cultural shift to online competition and gamification** – fueled by Peloton, Zwift, esports, and connected fitness trends, users are primed for digital competitive workouts. It also seems that for the average person, the pursuit of achievement is more enjoyable when it feels like a game - gamified habit

trackers, DuoLingo etc. Gamifying rowing could make it much more enticing for people who have not yet been sucked into the cult.

- **Mobile infrastructure** – modern app frameworks support low-latency, real-time multiplayer experiences at scale. Latency may have been a blocking factor 10 years ago.
 - **Esports Convergence** – Competitive gaming and competitive fitness are colliding (Zwift racing is now recognized by UCI Cycling). oArena makes rowing a spectator-ready digital sport, not just training.
 - **Gen Z & Social Fitness** – Younger athletes demand apps that are *mobile-native*, *social-first*, and *competitive*. oArena matches their lifestyle better than legacy training log software like ErgData.
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Stakeholders

- **Customers/Users (Rowers & Fitness Enthusiasts):**

- *Values:* **Competition**, motivation, fun, measurable progress, community.
- *Motivations:* To make solo workouts engaging, to improve fitness, to connect with others, and to prove your mettle in a competitive environment.
- *Behaviors:* For serious athletes: use the app every day either to log training or to compete. For casual users: join a live race a few times per month and do some solo training on the side.

- **Concept2:**

- *Values:* Brand reputation, ecosystem growth, hardware adoption.
- *Motivations:* Ensure the PM5 monitor remains the “gold standard” by supporting an ecosystem of apps. Maintain firmware to provide a seamless and intuitive user experience. Maintain position at the cutting edge of rowing hardware design.
- *Behaviors:* Maintain firmware openness, possible partnerships/licensing.

- **Gyms & CrossFit Boxes:**

- *Values:* Member retention, offering variety, supporting competitive culture, facilitating member growth.

- *Motivations:* Provide engaging challenges for members; showcase high-intensity rowing races; provide fun and nuanced interactive fitness experiences.
 - *Behaviors:* Host in-gym races, promote app usage among clients, adopt the C2 PM5 so more users come to their gym to use the app.
 - **Investors:**
 - *Values:* Market traction, subscription growth, differentiation from competitors, transition to online experiences from traditional practices.
 - *Motivations:* Capture connected-fitness market outside of high-cost hardware.
 - *Behaviors:* Provide early-stage funding, push for product adoption beyond rowers.
 - **Employees/Team:**
 - *Values:* Building an impactful product that redefines how rowers train, grow the sport of rowing.
 - *Motivations:* Career growth, equity, passion for fitness tech, care for the user.
 - *Behaviors:* Prioritize user experience, innovate around gamification, balance stability and growth.
 - **Regulators (Minor role at MVP stage):**
 - *Values:* Data protection, fair competition.
 - *Motivations:* Ensure user privacy, accurate marketing.
 - *Behaviors:* May intervene if the app handles personal health data at scale.
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User Journey (Customer / Athlete)

Persona:

- College rower, training remotely for summer. Wants structured, competitive workouts to stay motivated.

Journey:

1. Awareness & Onboarding

- Hears about oArena from a teammate's Instagram story showcasing a live race/victory.
- Downloads the app after a quick Google search, or by asking their friend for the link; connects via Bluetooth to their Concept2 erg in their garage.
- *Values tension:* Curious but skeptical — will this feel real, or just another leaderboard/workout logger? Will it be fun?

2. First Experience

- Starts with a quick “Just Row” session to test functionality.
- App instantly mirrors performance metrics and provides a fun and intuitive training experience with clean graphics which showcase the app's affinity for competition. Saves the workout upon completion.
- The simplicity builds trust — it works seamlessly, unlike some fitness tech that requires fiddling. User is now curious how an actual live race would feel.

3. First Competition

- Signs up for a 5x5 min interval workout hosted by another rower with a set start time.
- Joins the race lobby, sees avatars of competitors.
- The race time comes around; all participants are instructed to sit ready. At “Go,” the erg automatically syncs to race parameters, and they see real-time pacing comparisons with graphics which show relative competitor positions and pace.
- Adrenaline kicks in: it feels like lining up at the start line of a regatta. User is focused not only on their own performance (the typical hard workout paradigm), but how they are stacking up against the other competitors in real time.
- *Values tension:* Their need for fair competition depends on others rowing honestly — risk of people “sandbagging” or disconnecting mid-race. Also, if they are in a lobby with people much better or much worse than them (they either dominate or get crushed), they may immediately lose confidence in the value of the app for their personal training journey. It is important that users find and join

competitions where they stand a fair chance, and victory will be a challenge.

4. **Post-Workout Reflection**

- Results page shows finishing position, split times, and personal best markers compared with the competition.
- Sees metrics saved in history with a progress trend graph based on previous efforts.
- Option to share results on social media/Strava → dopamine hit, motivation reinforced.

5. **Habit Formation**

- Creates a recurring Wednesday “5k Challenge” and invites club teammates and friends. Other users can also freely join the event.
- Starts looking forward to these races; replaces boring solo rows with gamified sessions where high performance is awarded with prestige and social validation.
- Becomes part of a micro-community of online rowers.
- Starts logging all training (even low-intensity solo training) through the app for direct comparison with their community.

Values Tensions Highlighted:

- **Customer vs. Concept2:** oArena depends on Concept2’s continued support for open APIs; if Concept2 restricts access or makes sweeping firmware changes, users lose functionality.
- **Customer vs. Other Users:** The experience depends on fairness and integrity — cheating, quitting, poor competition, or connection issues undermine trust.
- **Customer vs. Gyms/CrossFit:** Athletes may prefer at-home privacy, while gyms push for in-person usage to drive community.